

Episcap: project log

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Living record. Read top to bottom to get up to speed. Figures regenerated from `results/`; see `docs/WORKFLOW.md`

This is the running record of the episcaf project: scaffolding antibody epitopes defined by 3D geometry into folded, assay-ready designs (RFDiffusion3 $\rightarrow$ ProteinMPNN $\rightarrow$ AlphaFold3). It is meant to be read in order, since each section is one iteration: what we tried, what we found, and what it changed for the next step. It is not a formal manuscript. Every figure is regenerated by a named script (recorded in `manuscript/figures/FIGURES.md`) and every number traces to `results/`; unconfirmed values are flagged [verify].

1 Motivation: scaffolding B-cell epitopes

1.1 Why scaffold epitopes

This project is about *B-cell epitopes*: the regions of an antigen that an antibody recognizes. Antibodies recognize these regions through their three-dimensional geometry, and many of the most interesting B-cell epitopes are discontinuous in sequence but contiguous in space, meaning the contacted residues are scattered along the antigen sequence yet fold together into a single surface patch. Linear methods cannot represent this. High-throughput serology platforms such as PepSeq tile an antigen into short, overlapping linear peptides, which works well for continuous, sequence-local epitopes. A linear peptide, however, cannot reconstitute a surface assembled from sequence-distant residues, and it discards the native conformation the antibody evolved to recognize.

The central question of this project is whether *scaffolding* an epitope produces a stronger, cleaner binding signal in the PepSeq assay than linear tiling does. By embedding the epitope residues in a folded protein, we can hold them in the conformation the antibody actually sees, and we can present distinct epitopes as separable, individually testable designs. If that conformational presentation matters, scaffolded designs should bind their cognate antibodies more reliably than the corresponding linear peptides. The longer-term goal is to extend PepSeq-style high-throughput assays beyond linear peptides to conformational, discontinuous epitopes.

1.2 The approach

We extract the epitope, meaning the residues an antibody contacts, from an antibody:antigen crystal structure. We embed those residues into a *de novo* protein scaffold with RFDiffusion, assign a sequence with ProteinMPNN, and then ask AlphaFold3 whether the design actually folds with the epitope held in its native geometry. Designs are kept or rejected on AlphaFold3 confidence and geometry-preservation metrics (Section 2). The full production pipeline, including the Nextflow orchestration, the RFDiffusion runs, and the AlphaFold3 evaluation, originates in Lawson Woods'

episcaf-experiments repository. Reproducing and then extending it is the starting point of this work.

1.3 Patches and islands

A single antibody:antigen complex can yield more than one epitope *patch*, labeled 0P, 1P, and so on, where each patch is a distinct cluster of antibody-contacting residues drawn from the same structure. A patch is in turn made of one or more *islands*, the spatially contiguous runs of epitope residues in three-dimensional space. This distinction matters because it predicts difficulty. Single-island patches can be enforced locally and tend to scaffold well, whereas multi-island patches require preserving the relative position and orientation of spatially separated regions, which is much harder. A concrete failure makes the point: PDB 2XQB, patch 0P, is a 56-residue epitope split across two separated islands, and across 2,624 RFdiffusion trials zero designs passed the downstream filters. Because this 0% rate holds over thousands of attempts, it looks like a real computational failure, not undersampling. The cause appears procedural: contigs (the constraints RFdiffusion is given) enforce epitope residue identity and ordering, but they do not strongly constrain the distances and orientations *between* islands, so the model can satisfy the contig and still lose the geometry an antibody needs.

1.4 Design pools and the DP4 questions

The project spans two PepSeq iterations. DP3 is the previous round: the known-antibody mAb set, where a solved antibody complex gives a ground truth for “good.” DP4 is the library we are now assembling — several thousand 103-mers that *mostly* re-target those same mAb epitopes (asking what a minimal epitope is and which design features predict binding) and, for a minority of antigens, extend to whole proteins with no known antibody. The no-antibody antigens are therefore one component of DP4, not the whole; but they are the component that forces the accessibility problem, since the antibody-clash filter cannot be computed without an antibody. Scoring therefore runs in two regimes:

- **Known antibody (all of DP3, most of DP4).** When the antigen has a solved antibody complex, we know the exact epitope and we have a ground truth for “good”: the design must hold the epitope in place and must not place scaffold mass where the antibody has to bind. This is the setting where we have already run designs through the assay and seen some success, and where the scaffolded-versus-linear question is most directly testable.
- **No antibody (one component of DP4).** For whole antigens with no solved antibody we do not know where the epitope is, so we tile the sequence into overlapping 12-residue windows (stepping by two residues) and scaffold each window, covering the antigen densely enough to catch whatever the real epitopes turn out to be. This is the polyclonal arm of DP4, and the one place the native-aware cylinder surrogate (Section 6) is needed, since there is no antibody to clash against.

DP4 is built to answer three questions, and they organize the rest of this log. (*Q1*) *What is a minimal epitope?*, how many islands and residues are needed for binding: whether an antibody needs one epitope island or both, and the island analysis (Section 8), and the per-island production run that tests it (Section 9), are aimed here. (*Q2*) *What predicts a successful design?*, which of epitope RMSD, scaffold RMSD, PAE, and accessibility actually track binding: this is what the decomposed metrics and composite scorer (Section 5) are for, and DP4 is built to answer it empirically by assaying designs sampled across that feature space and reading off what binds —

the non-circular experimental label we have been waiting for before fitting the scorer’s weights (Section 11). (*Q3*) *Can it extend to the polyclonal setting?*, tiling a whole antigen with no known antibody, where the cylinder surrogate (Section 6) stands in for the antibody-clash filter. These design ideas originate with the project’s earlier work; the scoring and metrics developed here are a refined evolution of them.

1.5 How the rest of this log reads

The log runs in two halves. The first establishes the method on the known-antibody (DP3) set and ends with the experimental ground truth that anchors it; the second uses that footing to design the next library, DP4. Each section motivates the next.

Part I — method and ground truth. We reproduce Lawson’s workflow and confirm our metrics match his (Section 2), tile antigens into assay constructs (Section 3), and compare the RFdiffusion1 and RFdiffusion3 design sets head to head (Section 4). We then rebuild scoring — the inherited filters bias selection toward rigid folds, so we decompose the metrics into a composite that picks the best designs per epitope (Section 5) — and recover the antibody-accessibility filter for targets with no antibody, using the native-aware cylinder surrogate (Section 6). The first experimental binding data then closes the loop (Section 7): it shows which metrics actually predict binding, and is what sets the scorer’s weights.

Part II — designing DP4. With the method grounded, we turn to the next library. We ask whether an antibody needs one epitope island or both (Section 8), build the per-island production run that lets the assay answer it (Section 9), and lay out the full set of DP4 components we are shipping (Section 10). Open questions and next steps close the log (Section 11).

2 The pipeline, and reproducing Lawson’s workflow

2.1 Pipeline

Each contig is carried through three stages. RFdiffusion3 generates 8 scaffold backbones holding the epitope segment fixed at its contig position (one GPU task per contig, `n_batches=1`). Each backbone is stripped to its $C\alpha$ /backbone trace and re-sequenced by ProteinMPNN, which is told to hold the epitope residues *fixed* — so it designs only the scaffold — and draws 8 sequences per backbone (temperature 0.1); the epitope’s fixed positions are read back from the contig, since an island sits at residues $N+1 \dots N+\ell$ of an N -flank contig. AlphaFold3 then predicts each sequence single-sequence (`--norun_data_pipeline`, one diffusion sample, seed 1), giving the $8 \times 8 = 64$ designs per contig that the four filters score against the designed structure. A small number of backbones fail to yield sequences in ProteinMPNN for reasons we have not yet diagnosed, so the realized design count falls marginally short of the nominal $5568V$ (for the $V = 20$ run, 111 344 of 111 360) [verify]. Generation code lives in `episcaf_pipeline/` and `scripts/`, and the metrics, scoring, and visualization code in `episcaf_analysis/`.

Benchmarking. Rough per-step throughput, to size cluster wall-time requests: RFdiffusion3 produces a contig’s 8 backbones in under 10 min on one A100; ProteinMPNN sequences ~ 215 backbones/hour (≈ 17 s for a backbone’s 8 sequences); AlphaFold3 single-sequence predictions are fast but unmeasured on our V100 nodes [verify]. A run’s wall-time then follows by multiplication — e.g. a 300-backbone ProteinMPNN batch is ≈ 1.4 h, for which we request 8 h to leave margin.

2.2 Reproducing Lawson’s baseline

Before trusting any new number we re-implement Lawson’s metrics and check them against his ground truth. `compute_metrics.py --validate` reads `dp2.parquet` (Lawson’s stored per-design values), recomputes `overall_rmsd` and `epitope_chunk_rmsd` from his ProteinMPNN PDBs and AlphaFold3 CIFs, and diffs them against the stored values. Agreement on this diff is the precondition for everything downstream: it confirms that our implementation of the four filters (Section 5.1) matches the workflow we are extending, so that later differences reflect the design protocol rather than a scoring bug.

3 Preprocessing: tiling antigens into assay constructs

The no-antibody tiling — the polyclonal component of DP4, not the whole of it — starts before any design is generated, by turning an antigen into a set of short peptide constructs to scaffold. Two things drive how we do this.

First, RFD3 needs an input crystal structure. The contig tells RFD3 which residues to hold fixed while it builds a scaffold around them, so those residues have to exist in a PDB. We therefore tile the *PDB-resolved* sequence (the residues actually present in the structure), not the deposited FASTA. 6M0J is the clearest case: its FASTA starts 14 residues before the crystal (chain E begins at residue 333, which is position 15 in the FASTA), so we drop those 14 and tile the 223-residue resolved region. 1D2K (392) and 4WAT (402) match their full chains.

Second, the constructs have to match the linear-epitope assay. Each tile is placed at the C-terminus of a constant 73-residue construct: a glycine-serine filler followed by the ENLYFQGA protease site. Because the prefix is constant, every member is synthesized at a constant length (103 residues for a 30-mer), and the protease site cleaves it down to the bare tile at the end, which is what the assay reads.

3.1 General tiling utility

The tiling step is a single utility (`episcap_pipeline/tile_fasta.py`) so we can point it at any target. It takes a sequence source (a FASTA, or the `antigen_seq` column of a ledger such as `dp2.parquet`), a window length, and a step, and writes the assay library in the DP2 annotated format (`library_member`, `sequence`, `category`, `model`, `designedSequence`, `designedSequenceLength`, `design_ID`, `target`). Windows are placed at the chosen step; when the last window stops short of the C-terminus, we add one final window so the C-terminal residues are still covered. We pinned the utility to what was already run: re-tiling 1D2K, 4WAT, and 6M0J at a 12-residue window and step 2 reproduces the existing 12-mer library exactly (191, 196, and 107 tiles, every tile and full construct identical).

3.2 Tiled 12-mers, and tiled 30-mers as linear controls

The existing set tiles three antigens into 12-mers. The next category is tiled 30-mers, which serve as the *linear controls*: unscaffolded constructs (model (`none`)) that present the bare peptide, exactly what the scaffolded designs are meant to beat. Because these controls are *not* scaffolded, they carry none of the constraint that ties the 12-mers to PDB-resolved residues: a linear peptide need not exist in any crystal, so we tile the full deposited **PDB FASTA** chain, which is gap-free, rather than the resolved `antigen_seq`. We tile all 59 mAb-target antigens plus the three tiled antigens at a 30-residue window and step 6.

The resolved sequence is gap-free but truncated; the FASTA fixes both. For each antigen we fetch the RCSB FASTA and identify the antigen chain by matching the resolved `antigen_seq` (`episcaf_pipeline/pdb_fasta.py`); the native window is the FASTA chain from where that match begins, which strips any N-terminal expression tag (restoring the same native start the earlier library used) while recovering the C-terminal residues the resolved sequence had dropped. Checking all 59 (`scripts/verify_antigen_seq_vs_pdb_fasta.py`, `results/antigen_seq_vs_fasta.csv`): the resolved `antigen_seq` is an exact contiguous slice of the FASTA for 58 of them, missing only terminal residues (up to 74); the lone exception, 6OKM, splices across an internal unresolved gap — a non-native junction. Tiling the FASTA recovers 535 residues overall and removes that fake junction, and the builder asserts every emitted tile is a substring of its source chain. The build (`episcaf_pipeline/build_dp4_tiled30mers_fasta.py`) reproduces the earlier library’s tiles where they overlap (e.g. 7OX3 `design_ID` 1 and 7 match exactly) and yields 2174 30-mers from the 62 antigens.

3.3 Why the design count is not (tiles $\times$ top- k)

A natural expectation is that taking the top k designs per epitope gives k times the number of tiles. It does not, and the gap is informative. Each tile is one epitope that RFD3 scaffolds into many designs (seeds $\times$ ProteinMPNN sequences), which are then AlphaFold3-validated and gated; we keep at most k per epitope, so the selected total is $\sum_{\text{epitopes}} \min(k, n_{\text{gated}})$. In practice the per-epitope yield is not the bottleneck: every epitope that survives has well over k designs (median ~ 190 scored). The shortfall is whole epitopes that drop out, because a tile whose residues are unresolved in the crystal cannot be aligned to the native antigen and is discarded at scoring (`crystal_epi_incomplete`). Table 1 shows this for top-5 selection.

antigen	tiles	epitopes lost (crystal)	epitopes kept	top-5 designs
1D2K	191	0	191	951
6M0J	107	15	92	450
4WAT	196	40	156	780

Table 1: Selected designs track (epitopes kept) $\times$ 5, not (tiles) $\times$ 5. 1D2K loses no epitopes; 6M0J and 4WAT lose 15 and 40 tiles to unresolved crystal regions. The small residuals (951 vs 955, 450 vs 460) are epitopes with fewer than 5 designs clearing the 2.5 Å gate.

4 RFD1 versus RFD3 on the same epitopes

With the metrics reproduced, we compared the two backbone generators head to head: the same DP3 epitopes run through RFDdiffusion1 (Lawson’s set) and RFDdiffusion3, each followed by ProteinMPNN and AlphaFold3, and scored against the identical four filters. We score over *every design generated*: a design missing any of the four filter metrics produced no scorable result — it failed somewhere in the pipeline or was dropped from Lawson’s deposited set — and counts as a non-pass rather than being excluded from the denominator. Lawson’s RFD1 set (`dp2.parquet`) is missing at least one metric (dominantly the AlphaFold3 result) for 26 870 of its 151 232 designs, far more than RFD3’s 12 of 150 948, so this convention penalizes RFD1 more, not less. On that basis RFD1+MPNN passes 840/151 232 (0.56%) and RFD3+MPNN 751/150 948 (0.50%) — still RFD1 modestly higher (restricting to the designs with all four metrics, the rates are 0.68% and 0.50%).

RFD3 is therefore at least an even trade for RFD1 on this task, which is the basis for standardizing on RFD3 going forward. The per-metric distributions are comparable (Fig. 1), though RFD3 puts a bit more mass at low epitope and overall RMSD.

We are really comparing two sequence distributions. Each protocol samples sequences from a backbone-conditioned distribution $p(s | b)$ and is judged by the same pass rate (Eq. 1), so a difference in $\hat{P}_{\text{pass}}$ comes from a difference in those distributions over comparable backbones. ProteinMPNN optimizes $p(s | b)$ directly through its inverse-folding objective, while RFD3 produces it only as a by-product of all-atom denoising. The two landing this close is the result worth keeping.

RFD1 vs RFD3 on the same DP3 epitopes (both + ProteinMPNN + AlphaFold3); distributions density-normalized over designs with all four metrics
four-filter pass rate per design generated: RFD1+MPNN 840/151,232 = 0.56% RFD3+MPNN 751/150,948 = 0.50% (designs missing ≥ 1 metric count as non-pass: RFD1 26,870, RFD3 12; over valid only 0.68% vs 0.50%)

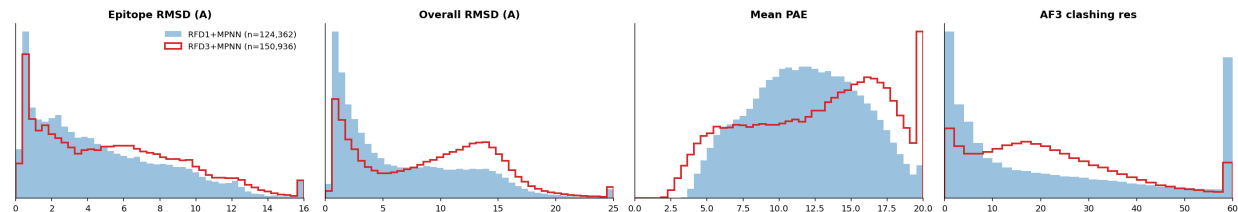


Figure 1: RFD1+MPNN (blue, filled) versus RFD3+MPNN (red, outline) on the same DP3 epitopes, across epitope RMSD, overall RMSD, mean PAE, and AF3 clashing residues, density-normalized over designs with an AF3 result (per-trial n in the legend). Pass rates and the missing-prediction asymmetry are noted in the title. Regenerated by `episcaf_analysis/viz/plot_rfd1_vs_rfd3.py`.

The aggregate pass counts hide a more interesting difference in *where* the passes land (Fig. 2). RFD1’s passes are highly concentrated: a single target (7ox3_OP) accounts for 630 of its 840 passes, and just two targets account for the large majority. RFD3 spreads its 751 passes across more targets (16 with at least one pass, versus 10 for RFD1), and its per-target leaders are different (4xwo_5P, 8pww_OP, and 8db4_OP all pass well under RFD3 but barely or not at all under RFD1). For a library meant to cover many epitopes, that broader coverage is more useful than a similar total piled onto one target.

5 How we score a design — and why we changed it

Once a design is generated it has to be judged. We inherited Lawson’s four-filter ground truth, found that two of its filters bias selection toward rigid, well-ordered folds, and rebuilt scoring around decomposed, epitope-specific metrics combined in a single composite. This section follows that arc: the inherited filters, why they mislead, the decomposition that fixes it, and the composite scorer that ranks designs and picks the best per epitope.

5.1 Four-filter ground truth (DP3 set)

The DP3 set itself is a 59-epitope subset of the 1134 cleaned antibody–antigen complexes in the AbDb directory (`configs/paths.py`, `ABDB_CLEANED_PDB_DIR`). The selection condition — which every one of the 59 satisfies in `dp2.parquet` — is at most two epitope islands (13 single-island, 46 dual-island) and a parent antigen longer than 103 residues (the antigen lengths run 105–909); the other 1075 complexes are the candidates that condition drops [verify].

A design passes when all four conditions hold: overall backbone RMSD ≤ 2 Å, epitope-chunk RMSD ≤ 1 Å, mean predicted aligned error < 5 , and zero antibody residues within 4 Å of non-

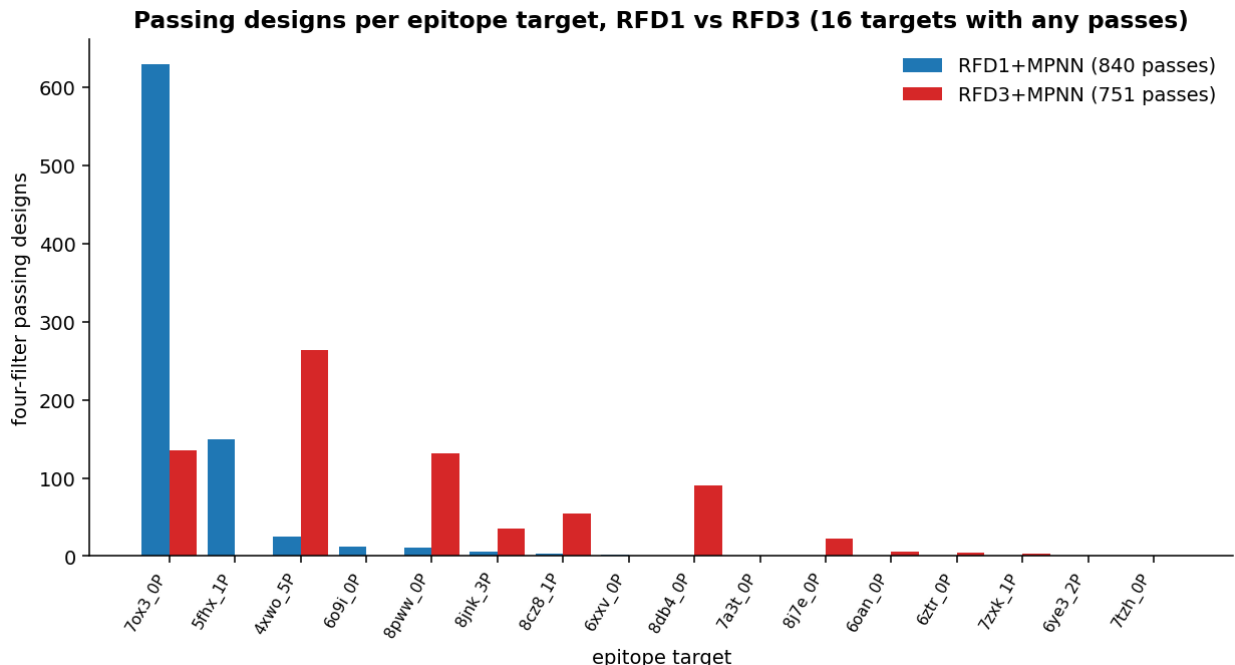


Figure 2: Four-filter passing designs per epitope target, RFD1+MPNN (blue) versus RFD3+MPNN (red). RFD1 is dominated by 7ox3_OP; RFD3 distributes passes across more targets. Regenerated by `episcaf_analysis/viz/plot_passes_per_epitope.py`.

epitope scaffold residues (`af3_n_clash_res`). Clash detection Kabsch-aligns the AlphaFold3 epitope onto the true epitope, then checks non-epitope scaffold heavy atoms against antibody atoms. The antibody coordinates always come from the AbDb crystal structure, not from AlphaFold3.

We summarize a design protocol by its empirical pass rate. If a protocol produces N designs s_i and $f(s_i) = 1$ when design i clears all four filters (and 0 otherwise),

$$\hat{P}_{\text{pass}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[f(s_i) = 1]. \quad (1)$$

This is just the fraction of designs that pass, and it is the quantity we compare between protocols and report per epitope.

5.2 The filters bias toward rigid folds

The first thing we changed was the filtering. The original filters score a design with two global continuous metrics, overall RMSD and mean PAE, that treat the epitope and the scaffold as one unit. We inherited this from standard de novo practice, but it biases selection toward globally rigid, well-ordered folds and quietly rejects designs whose scaffold is flexible while the epitope is well positioned (or the reverse). Splitting each metric into epitope- and scaffold-specific terms (Section 5.3) is what lets us see, and recover, those designs.

To compare design sets on a common footing we use epitope-chunk RMSD $< 2.5 \text{ \AA}$ as a single bar. All three tiled 12-mer antigens pass it more often than the pooled DP3 set:

set	pass rate ($< 2.5 \text{ \AA}$)	median epitope RMSD ($\text{\AA}$)
DP3 / mAb (pooled)	30.0%	5.00
6M0J	47.0%	2.65
1D2K	60.2%	1.85
4WAT	84.4%	0.43

Even the hardest 12-mer antigen (6M0J) clears the bar more often than DP3. Figure 3 shows the full six-metric comparison.

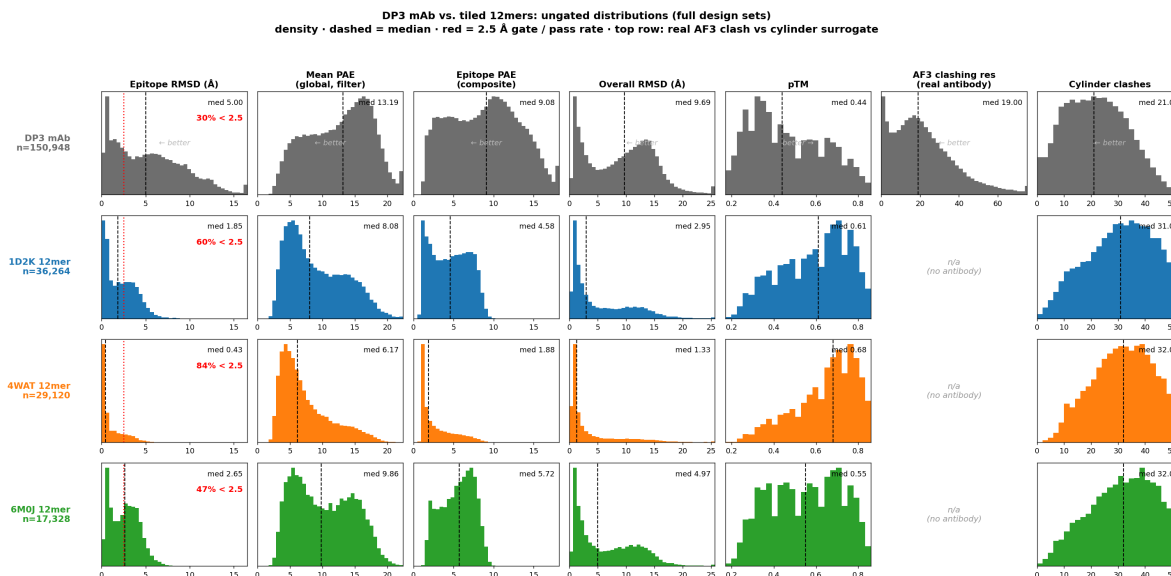


Figure 3: Ungated metric distributions (full design sets), DP3 mAb versus the three 12-mer antigens, across epitope RMSD, global mean PAE (the four-filter metric), epitope PAE (the decomposed metric the composite scores on), overall RMSD, pTM, AF3 clashing residues (antibody set only), and cylinder clashes (the plain count `cylinder_ca_clashes`, not the native-aware carve). Dashed line marks the median; red marks the 2.5 Å epitope-RMSD gate with the per-set pass rate. Regenerated by `notebooks/dp3_vs_12mer_distributions.ipynb`.

Caveats (important for interpretation). This comparison is not yet apples-to-apples. The 12-mer epitope is a single contiguous 12-residue stretch, while DP3 epitopes are larger and often multi-island, so 12-mer RMSD and PAE are mechanically lower in part by definition, and the pooled DP3 30% averages over heterogeneous epitopes. The clean control is to stratify within DP3 by island count and size (Section 8). The cylinder counts are also not comparable across sets as absolute numbers. The 12-mer medians (~ 31 – 32) run higher than DP3 (~ 21) mainly because the designs are a fixed length (~ 104 residues) and the 12-mer epitope is small, which leaves ~ 91 non-epitope scaffold residues whose $C\alpha$ atoms can land in a fixed cylinder against fewer for DP3’s larger epitopes. That is a scaffold-mass effect, not more real clashing, which is why we apply the native-aware carve and rank the cylinder per antigen instead of against one absolute cutoff (Section 6). As a check, the DP3 cylinder median (≈ 21) tracks its real `af3_n_clash_res` median (≈ 19).

Global versus epitope PAE. The two PAE columns show why we decompose, and they are easy to confuse. The four-filter (and Lawson’s original) uses the *global* mean PAE (< 5); on this

DP3 set it sits high, a median of 13.2 with only $\sim 7\%$ of designs below 5. The scaffold region drags that value up, since AlphaFold3 predicts it less confidently, while the epitope itself is much better resolved (epitope PAE median 9.1, $\sim 23\%$ below 5). So a design can hold its epitope well and still fail the global PAE filter on scaffold uncertainty that need not matter for binding. That is the bias the decomposed metrics relax, and it is why we show both columns and score the epitope-specific PAE rather than the global one.

5.3 Decomposed metrics

The two continuous filters, overall RMSD and mean PAE, treat the epitope and the scaffold as a single unit. We split each into epitope- and scaffold-specific terms (epitope RMSD, scaffold RMSD, epitope PAE, scaffold PAE), retaining the zero-clash hard filter throughout. Decomposition lets selection reward a well-placed epitope on a flexible scaffold (or the reverse) instead of demanding global rigidity, and it lets us ask whether the optimal rigidity profile is epitope-dependent rather than universal. Selection over the four metrics is organized as a 2×2 of strict and relaxed thresholds on (epitope, scaffold), with a flexible/flexible arm as a negative control.

5.4 Composite scorer and per-epitope selection

Decomposing the metrics raises the next question: how to combine them into one ranking and take the best designs per epitope. The scorer (`episcaf_analysis/score.py`) gives each design a weighted sum of per-metric scores,

$$S(d) = \sum_m w_m \varphi_m(x_m(d)), \quad (2)$$

where each activation φ_m maps a raw metric to a higher-is-better score: either a *percentile* within the scored population (rank-based, cohort-dependent) or a *sigmoid*

$$\varphi(x) = \frac{1}{1 + \exp(s k (x - x_0))}, \quad s = +1 \text{ if lower is better (else } -1), \quad (3)$$

an absolute, saturating score with a tunable midpoint x_0 and steepness k . This is exactly a single-layer perceptron with hand-set weights — the activations are the per-metric nonlinearities, the w_m are the weights — so “fitting” it later (Section 11) means learning the w_m (and x_0, k) from data. The dials live in `episcaf_analysis/presets.py`:

metric	weight
accessibility	0.35
epitope-chunk RMSD	0.35
overall RMSD	0.15
epitope PAE	0.15

These weights are not arbitrary: they are set from the DP3 binding experiment (Section 7), which finds that accessibility and epitope RMSD are the two metrics that actually track binding, so they carry the weight and overall RMSD and PAE are kept small. The accessibility term itself depends on the input — the *real* clash `af3_n_clash_res` when a known antibody is available, the native-aware cylinder surrogate (Section 6) when it is not — at the same weight either way.

We then **rank, not gate**: transform each metric within its scope (per antigen for the 12-mer set, pooled for the mAb set), weight-sum, and keep the top five per group. We deliberately

apply no hard threshold first — it would only discard designs the ranking already buries, and risks dropping one that is weak on a single axis but excellent overall, which the soft per-metric penalties handle more gracefully. (A gate is still available as an optional pre-filter, used in the diagnostics of Section 6, just not in picking the designs we ship.) Two choices remain open and are carried to Section 11: fitting the weights against binding rather than hand-setting them, and moving the percentile activation to the population-independent sigmoid.

6 Accessibility without an antibody: the cylinder surrogate

One of the strictest filters in the DP3 setting concerns antibody accessibility: we place the real antibody where it sits in the crystal complex and reject any design with scaffold mass in its path. That filter is only available where a real antibody exists (all of DP3, most of DP4). Most targets, though, have no solved antibody — the polyclonal arm of DP4 tiles whole antigens with no known binder (Section 1) — so there is nothing to clash against. We recover the filter with a native-antigen-aware cylinder surrogate: a cylinder fitted to the epitope plane along the antibody-approach normal that counts non-epitope scaffold mass intruding into the volume an approaching antibody would need (Fig. 4).

6.1 Definition

With no antibody available, we approximate the volume an antibody would occupy with a cylinder fitted to the epitope. The epitope $C\alpha$ coordinates $\{x_j\}_{j=1}^m$ define a plane: the centroid is $c = \frac{1}{m} \sum_j x_j$, and the plane normal $\hat{n}$ is the smallest-variance singular vector of the centered coordinates $\{x_j - c\}$ (the antibody approach direction), oriented to point away from the antigen body. The cylinder base is $b = c + d_0 \hat{n}$ with offset $d_0 = -4 \text{ \AA}$. A scaffold $C\alpha$ at position p lies inside the cylinder when, writing $v = p - b$, its axial and radial coordinates satisfy

$$0 \leq v \cdot \hat{n} \leq H \quad \text{and} \quad \|v - (v \cdot \hat{n}) \hat{n}\| \leq R, \quad (4)$$

with height $H = 40 \text{ \AA}$ and radius $R = 16 \text{ \AA}$. The plain count is the number of non-epitope scaffold $C\alpha$ satisfying (4); the native-aware count additionally drops any such atom within $1 \text{ \AA}$ of a native-antigen heavy atom, i.e. sitting where the native antigen already sits, which a real antibody tolerates. The geometry core is `episcaf_analysis/native_cylinder_core.py` (self-tested).

6.2 The native-aware carve removes false penalties

A naive cylinder over-penalizes, because some scaffold mass sits exactly where the native antigen already sits, a volume a real antibody clearly tolerates since it binds the native antigen there. The native-aware carve fixes this: among truly clash-free designs, we drop scaffold $C\alpha$ atoms within $1 \text{ \AA}$ of a native-antigen heavy atom. Of the 3186 DP3 designs with zero true antibody clash, the number the plain cylinder flags as heavily penalized (count ≥ 10) falls from 1112 to 384 once the native footprint is carved out, so **728 false-positive penalties are removed**, while genuinely clashing designs are left essentially untouched (Figs. 5 and 6).

A pipeline constraint worth recording: crystal gaps. We measure 12-mer epitope fidelity by aligning the AlphaFold3 epitope onto the native crystal epitope, matched by residue number (`build_12mer_metrics.py`). When a tile’s epitope residues are unresolved in the crystal (a gap in the deposited structure), the native $C\alpha$ anchors are missing, the Kabsch alignment cannot be

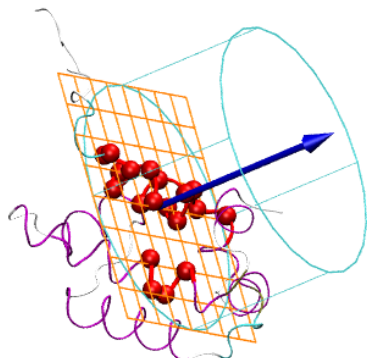


Figure 4: The accessibility cylinder. Epitope residues (red) sit on the scaffold (magenta); the epitope C α atoms define a plane (orange grid) whose outward normal (blue arrow, the antibody-approach direction) orients the cylinder (cyan). Scaffold C α atoms inside the cylinder count against accessibility. Rendered in VMD by `known_antigen/analysis/cylinder_vis/visualize_cylinder.tcl`.

formed, and the tile is dropped with status `crystal_epi_incomplete`. So which 12-mer epitopes are scorable at all depends on crystal coverage: an antigen with gaps over the tiled region contributes fewer (or no) scorable tiles, regardless of design quality. This is exactly what sets the epitopes-kept counts in Table 1 (6M0J and 4WAT lose 15 and 40 tiles), and the same constraint reappears in the per-island production run (Section 9.2).

6.3 Interpreting the cylinder: where passing designs land

The cylinder is easiest to read by overlaying, for each metric, the full design population against the designs we would actually advance (Fig. 7). For DP3 the advanced set is the four-filter passes (751 designs); for DP4 it is the top three per epitope by composite score. On every geometry and confidence metric the passing designs concentrate where we expect: low RMSD, low PAE, high pTM. The two cylinder columns are the interesting ones. Under the plain count the passing DP3 designs, which we know the antibody accepts, sit low (median 10), while the DP4 top-3 sit higher (medians 24–31). This is the scaffold-mass effect again, not a sign the DP4 designs are worse: with a 12-residue epitope on a ~ 104 -residue design, more non-epitope scaffold mass falls in the fixed cylinder even among the designs we would pick. The native-aware carve directly counteracts this: it lowers the passing medians to 6 (DP3) and 14–23 (DP4), pulling the larger DP4 scaffolds down most, because much of their extra in-cylinder mass coincides with where the native antigen sits. This is why we lean on the native-aware carve and per-antigen scoring rather than one absolute cutoff, and it gives us a concrete reference for a tolerable cylinder count, namely the DP3 passing range.

The cylinder is a good stand-in for the real filter. On DP3, where the true antibody clash is

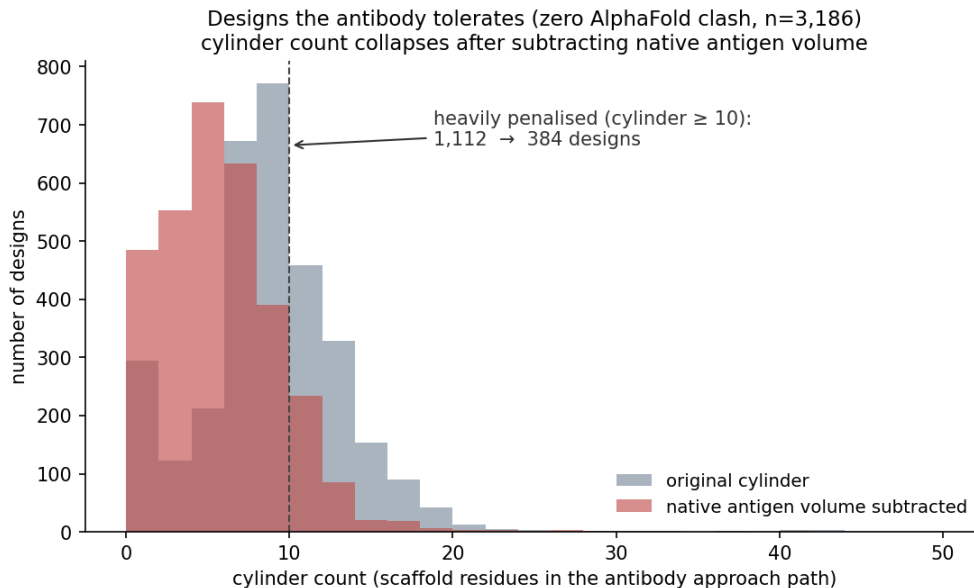


Figure 5: Cylinder count among clash-free DP3 designs (zero AlphaFold3 antibody clash, $n = 3186$), before (grey) and after (red) subtracting the native-antigen volume. The heavily penalized tail (count ≥ 10) collapses from 1112 to 384. Regenerated by `episcaf_analysis/viz/plot_fp_reduction.py --native metrics_native_cyl.csv --mode dist`.

known, it predicts clash-free designs (`af3_n_clash_res=0`) with AUC 0.93 (`cylinder_native_aware` 0.935, a little better than the plain count 0.924; clash-free versus clashing medians of 5 versus 22). So it stands in well for the filter we lose on DP4, and the native-aware carve helps the ranking, not just the picture (`episcaf_analysis/cylinder_surrogate_auc.py`). The same ordering survives without the well-predicted gate: recomputed over the full DP3 set (147 736 designs, `scripts/dp3_native_cylinder.py`), the native-aware count correlates with the true antibody-clash count more strongly than the plain count (Pearson 0.38 vs 0.34). This correlation is measured on the *well-predicted* designs (epitope-chunk RMSD $< 2.5 \text{ \AA}$): a design AlphaFold3 folds poorly carries an unreliable epitope placement, which would only add noise to a clash-versus-surrogate comparison, so it is excluded *from this statistic*. Selecting the best designs per epitope is a different task and applies no such hard threshold — it ranks on the composite score directly (Section 5.4), which is why the per-island selection (Section 9.3) does not gate at all and simply ranks.

The AUC here shows the cylinder reproduces the *in-silico* clash it was built to approximate. The stronger test — whether it tracks real *binding* — is the one the DP3 assay lets us run, and it is where the cylinder turns out to be the best predictor we have; we take that up next (Section 7.4), since it is also what sets the cylinder’s weight in the scorer.

7 Experimental binding data: the first ground-truth label

Everything up to this point has been scored *in silico*: the four filters and the composite rank designs by how AlphaFold3 sees them, with no measurement of whether a design actually binds. We now have the first experimental readout for the DP3 (known-antibody) set. This is the non-circular label the scorer has been waiting for (Section 11), so this section sets out exactly what the data is,

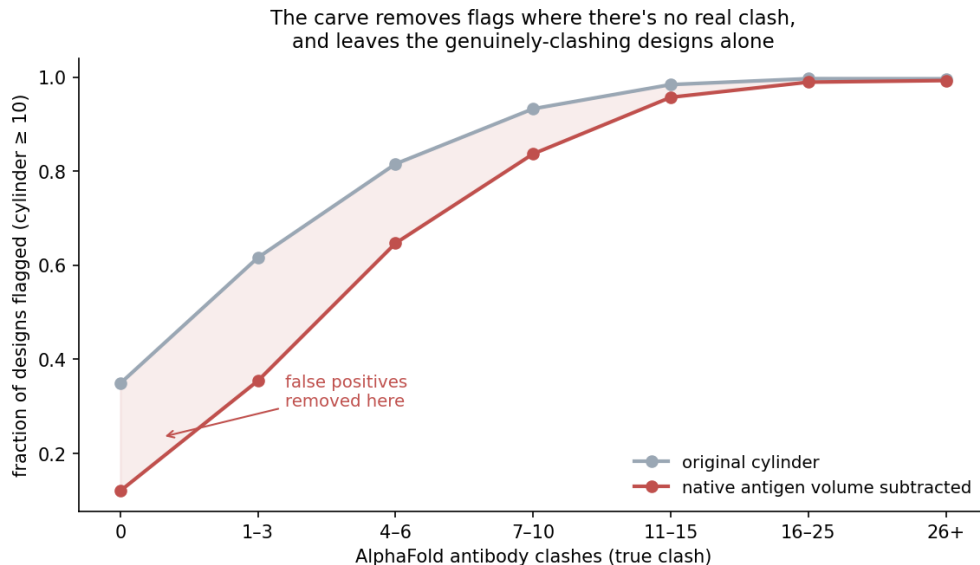


Figure 6: The carve removes flags selectively. Fraction of designs flagged (cylinder ≥ 10) across bins of true antibody clash, before versus after the carve. The gap is large where there is no real clash (false positives removed) and closes where the clash is real (true signal retained). --mode selectivity.

what it looks like, and — importantly — how far it can be trusted, before any weight is fit to it.

7.1 How a binding measurement is made

The DP3 designs were carried into the PepSeq assay. Each library member is displayed as its constant-length construct (the 103-mer of Section 3), and the assay reads out, by sequencing, how much of each member is pulled down by a given monoclonal antibody relative to a no-antibody control (NoAb) run on the same plate. The working readout is the one used throughout: $\log_{10}(1 + \text{Ab})$ against $\log_{10}(1 + \text{NoAb})$, with the designs cognate to that antibody — those scaffolding its epitope — highlighted. A design that binds sits above the $y = x$ diagonal: more pulldown with the antibody present than without. We summarize each design by its *cognate log-enrichment*, $\log_{10}(1 + \text{Ab}) - \log_{10}(1 + \text{NoAb})$ for the antibody raised against its target.

The measurements arrived as two assay runs, and the run parameters are not identical: the six IM226 antibodies were read at 0.1 pmol with antibody-first bead capture, while the two IM229 antibodies were read at 1 pmol (one, 8db4, at a ten-fold higher antibody concentration) with a different bead order. Absolute counts are therefore not comparable across runs or even across antibodies, which is the first reason the analysis below stays *within* an antibody rather than pooling. The full wet-lab protocol is recorded with the experimental methods; here we treat the intensities as provided.

7.2 What we have

Eight monoclonal antibodies, the subset of the original ten that assayed cleanly: 4xwo was dropped (its synthesized antibody was too low-yield) and 7a3t was dropped (its epitope was too small for RMSD to discriminate). Their designs remain in the library but have no usable antibody column. The two runs share the same 1000 DP2 library members; 403 are our scaffolded epitope designs

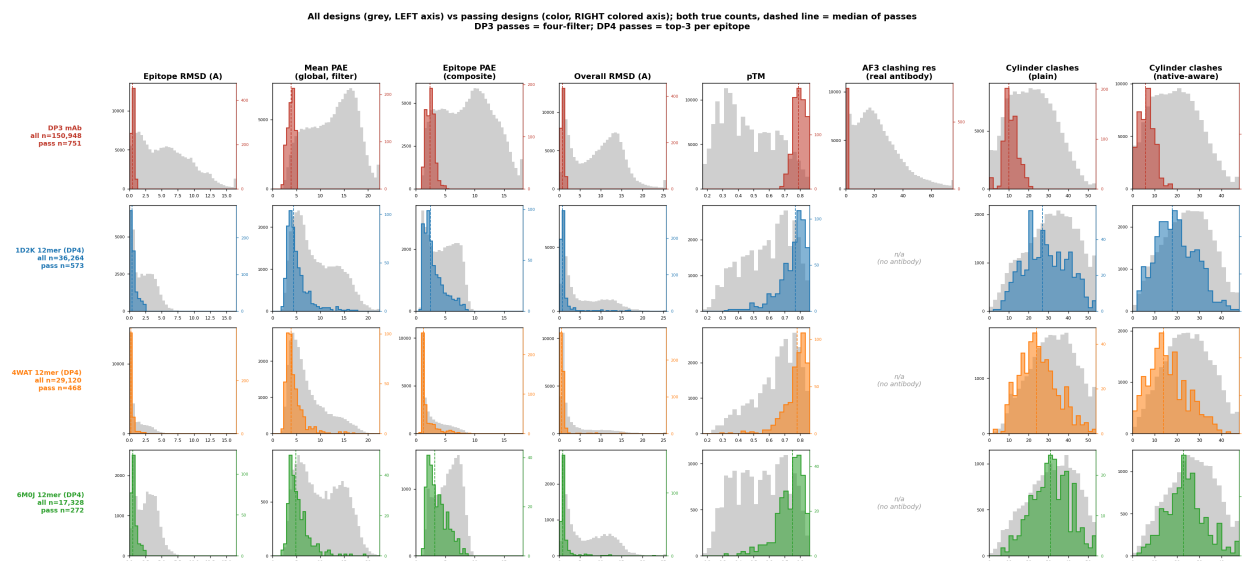


Figure 7: All designs (grey, left axis) and passing designs (color, right colored axis), both in true counts, per metric and per row; the dashed line marks the median of the passes. The twin axes keep the full population and the much smaller passing set each at its own scale, so the shape of the accepted distribution and where it lands are both visible. DP3 passes are the four-filter set; DP4 passes are the top 3 per epitope by composite score. The two rightmost columns show the accessibility surrogate both ways — the *plain* $C\alpha$ count (`cylinder_ca_clashes`) and the *native-aware* curve (`cylinder_native_aware`), which drops scaffold mass coincident with the native antigen. The curve shifts the passing medians down (10→6 for DP3, 24–31→14–23 for DP4), most for the larger DP4 scaffolds. Regenerated by `episcaf_analysis/viz/plot_passes_overlay.py`.

(`category=scaffoldedAbEpitope`) and the rest are control content (pMHC and published binders, which also carry the only measured K_d values — none of our designs do). Of the 403, 377 target one of the eight usable antibodies; the remaining 26 target the two dropped ones.

Each design’s scaffold sequence matches `scaffolded_epitope_seq` in `dp2.parquet` one-to-one (403 of 403), so we can attach its AlphaFold3 metrics directly and ask how binding relates to them (`scripts/build_dp3_binding_join.py` → `results/dp3_binding_metrics.csv`; figure and numbers from `scripts/analyze_dp3_binding.py`). Figure 8 shows the cognate designs against the library background for each antibody; Table 2 gives the per-antibody counts and median enrichments.

7.3 How reliable it is, and how far it goes

The headline is encouraging: across all eight antibodies the scaffolded designs enrich over the no-antibody baseline, almost all of them above the diagonal, and for the two well-sampled antibodies this holds for the great majority of designs. The scaffolding approach produces binders. That is a real, positive result.

But three properties bound how much this data can teach about *which* design features predict binding, and they all point the same way — toward a starting prior, not a fit.

Everything assayed already passed. The library was selected for synthesis from the top of the four-filter ranking, so every one of the 377 cognate designs has `is_pass` = true. The metrics consequently span only their passing range (`overall_rmsd` 0.35 to 1.95, `epitope_chunk_rmsd` 0.22

DP3 binding: cognate scaffolded designs (red) vs library background (grey)

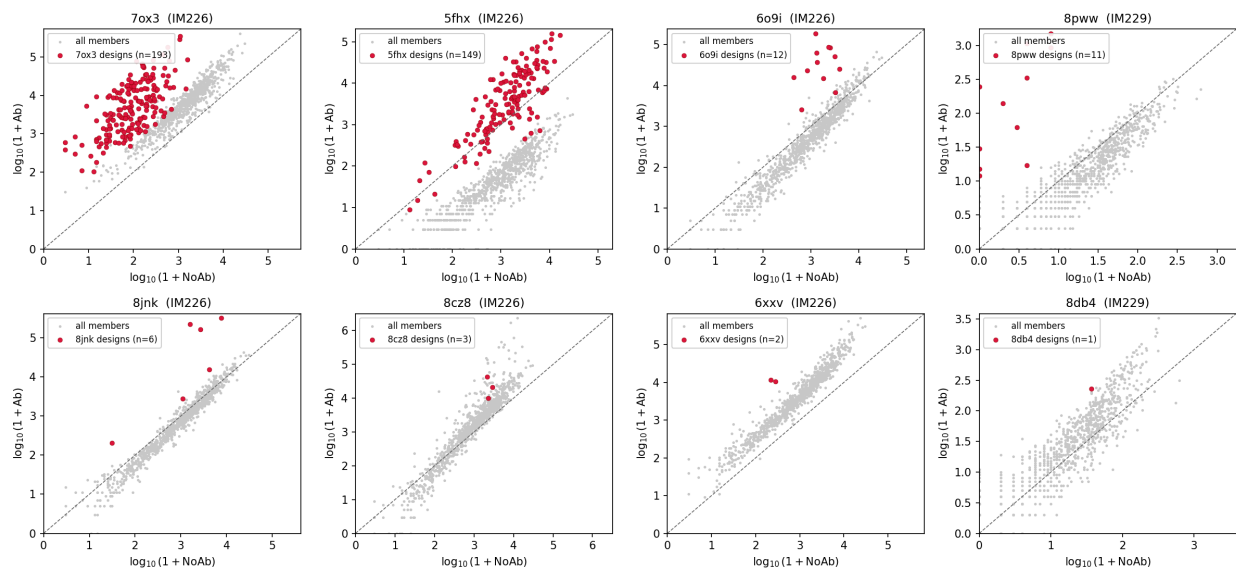


Figure 8: DP3 binding, one panel per antibody: $\log_{10}(1 + \text{NoAb})$ (x) versus $\log_{10}(1 + \text{Ab})$ (y), all 1000 library members in grey, the cognate scaffolded designs in red, dashed line $y = x$. The two well-sampled antibodies (7ox3, $n = 193$; 5fhx, $n = 149$) put the bulk of their designs clearly above the diagonal; the small- n antibodies show a handful of points each, mostly above the line. Panels are on independent axes because absolute counts are not comparable across antibodies/runs.

to 0.95, mean_pae 3.0 to 5.0, and zero antibody clash by construction). There is no failing design in the set to contrast against, so the filter itself has no variance here and cannot be tested for whether it separates binders from non-binders.

The pooled signal is a between-antibody artifact. Pooling all 377 designs, lower epitope RMSD tracks higher enrichment (Spearman $\rho = -0.43$), which looks like exactly the predictor we want. It is mostly an illusion of pooling: different antibodies sit at different baseline enrichments *and* different metric distributions, so a correlation across the pool largely reflects which antibody a design belongs to. Within a single antibody the relationship nearly vanishes (epitope RMSD versus enrichment: $\rho = -0.20$ for 7ox3, -0.07 for 5fhx), and the other metrics are similarly weak and inconsistent in sign.

The sample is two antibodies deep. 7ox3 and 5fhx supply 342 of the 377 designs; the remaining six antibodies have between one and twelve each. A within-antibody fit is only meaningfully possible for two epitopes, which is far too narrow to learn weights that would generalize across targets.

7.4 What does predict binding: the accessibility cylinder

The deconfounding move above — comparing designs *within* an antibody rather than across the pool — is also how we find the one metric that does carry signal. For each filter metric we regress cognate log-enrichment on it two ways: pooled across all 377 designs, and after removing each antibody’s own mean from both the metric and the enrichment (a fixed-effect centering that strips the baseline offsets the pooled view conflates). The contrast between the two numbers is the test: a predictor that is real survives the centering; an artifact collapses. Figure 9 shows both views for every metric, and we add the *accessibility cylinder* (Section 6) computed on the assayed designs’

antibody	run	cognate designs	median log-enrichment	fraction above baseline
7ox3	IM226	193	1.68	1.00
5fhx	IM226	149	0.42	0.81
6o9i	IM226	12	1.44	1.00
8pww	IM229	11	1.85	1.00
8jnk	IM226	6	1.23	1.00
8cz8	IM226	3	0.86	1.00
6xxv	IM226	2	1.65	1.00
8db4	IM229	1	0.79	1.00

Table 2: Per-antibody cognate scaffolded designs. “Above baseline” = log-enrichment > 0 (above the NoAb diagonal). Two antibodies (7ox3, 5fhx) account for 342 of the 377 designs; the other six carry 35 between them.

own AlphaFold3 structures (`scripts/assayed_native_cylinder.py`).

metric	pooled r	within-antibody r	survives deconfounding?
cylinder, native-aware	−0.23	−0.22 ($p=2 \times 10^{-5}$)	yes — essentially intact
cylinder, plain	−0.24	−0.18	mostly
epitope RMSD	−0.41	−0.14	no — mostly between-antibody
overall RMSD	−0.17	−0.08	no
mean PAE	−0.02	+0.07	no (and wrong sign)

Table 3: Pearson correlation of each metric with cognate log-enrichment, pooled versus within-antibody (fixed-effect centered). The accessibility cylinder is the only metric whose correlation barely changes between the two — it is a genuine within-antibody predictor, not a pooling artifact — and the native-aware carve is stronger than the plain count.

Three things make the cylinder result stand out (Table 3). First, it is the only metric that survives deconfounding: epitope RMSD looks like the best predictor when pooled ($r = -0.41$) but collapses to -0.14 once the antibody offset is removed, whereas the cylinder holds from -0.23 to -0.22 , with the same negative sign within all three well-sampled antibodies (7ox3 -0.24 , 5fhx -0.18 , 6o9i -0.61). Second, the sign is the mechanistically expected one and matches the surrogate’s purpose: scaffold mass placed where the antibody must approach predicts *less* binding. Third — and this is why the cylinder can show a signal where the filter metrics cannot — the cylinder was never part of the four-filter, so it kept its full dynamic range in the assayed set (native-aware count 0 to 17, median 6) while the filter metrics were squeezed into their passing slivers. The native-aware carve, which drops scaffold $C\alpha$ atoms that merely reoccupy the native antigen’s own volume, correlates more strongly than the plain count (-0.22 vs -0.18), a direct experimental endorsement of the exact refinement we apply where there is no antibody.

This is the first time the cylinder is validated against *experimental* binding rather than against the in-silico true clash it was built to approximate (the AUC of Section 6). The two lines of evidence agree, and they upgrade the starting prior: of everything we measure, the accessibility surrogate — the metric we most need for the no-antibody designs, where it is the only accessibility signal available — is the one DP3 binding most supports.

This is what sets the scorer’s weights (Section 5.4): accessibility and epitope RMSD carry

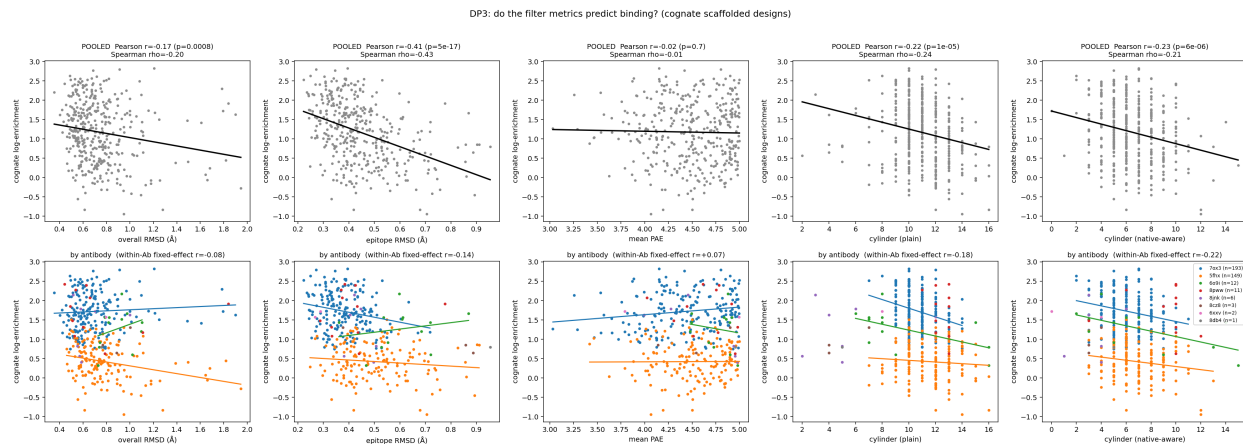


Figure 9: Each metric versus cognate log-enrichment. Top row pooled (one trend line); bottom row colored by antibody with per-antibody trend lines and the within-antibody fixed-effect correlation in the title. Epitope RMSD’s strong pooled slope (column 2) flattens once colored by antibody — the pooled line was riding the gap between the high-binding 7ox3 band and the low-binding 5flx band. The native-aware cylinder (right column) slopes down both pooled and within *every* antibody: more scaffold mass blocking the antibody approach, less binding.

the weight because they are the two metrics that track binding, and overall RMSD and PAE are kept small because they do not. What DP3 cannot yet do is *fit* the full composite, because the data has no failing designs and only two antibodies are sampled deeply — so the weights stay a hand-set prior, not a fit. Closing that gap is the explicit purpose of the DP4 metric-space sampling (Section 11): deliberately assay designs spread across the metric space, including ones the current filters would reject, so the weights can finally be fit with the variance and breadth this first set lacks.

8 Does binding depend on one epitope island or both?

A two-island epitope binds its antibody through two spatially separate patches, and the native structure alone cannot say how the interaction is divided between them. Does the antibody need island A, island B, or only both together? The crystal complex shows the bound state but not which contacts are load-bearing, and no purely computational metric on the native geometry can answer it. The way to find out is to build the reagents that separate the two contributions: scaffold island A on its own, island B on its own, and the two in their native arrangement, then let the PepSeq assay read off which presentations the antibody still binds. If a single island suffices, it carries the interaction; if neither binds alone, the epitope is genuinely discontinuous and both islands are required. Different two-island epitopes may distribute their binding differently, and that distribution is exactly what the per-island run (Section 9) is built to measure.

A related, computational question: are two islands harder to scaffold? Before turning to binding, the design data answer a narrower question on their own: holding the native geometry fixed, are two-island epitopes harder to scaffold than one? This is the island-*count* control, a first cut at the within-DP3 stratification of Section 11. We split the DP3 design set (RFD1+MPNN, `dp2.parquet`) by epitope island count, where an island is a spatially contiguous patch of epitope residues and the count is the number of fixed segments in the contig (the `epitope_chunks` field).

Of the 59 epitopes, 13 are single-island and 46 are two-island; none have more under this grouping. Restricting to designs with an AF3 result, that is 549 single-island and 123 813 two-island designs.

Across every four-filter metric the single-island designs sit at better values (Fig. 10): median epitope-chunk RMSD 2.22 vs 3.86 Å, median global PAE 9.5 vs 11.9, and fewer AF3 clashes. The binary pass rate runs the other way — single-island 0.24% (2/832) vs two-island 0.56% (838/150 400), on the same per-design denominator as Section 4 — but that is misleading: of the 838 two-island passes, 630 come from a single easy outlier (7ox3_OP) and another 149 from 5fhx_1P, so two epitopes account for 779 of them. Strip the outlier and the two-island rate falls below single-island (0.17% vs 0.36% per valid prediction); the single-island set has no comparable outlier and is broadly easier. So on the distributions, fewer islands is easier to scaffold, consistent with the multi-island difficulty argument in Section 1; the binary rate just hides it. What remains for the clean control is to split further by island *size*, since the single-island epitopes are also shorter on average.

So the two questions point the same way for the design itself — multi-island is both biologically ambiguous and harder to scaffold cleanly — and both motivate building each island as a standalone, individually testable construct.

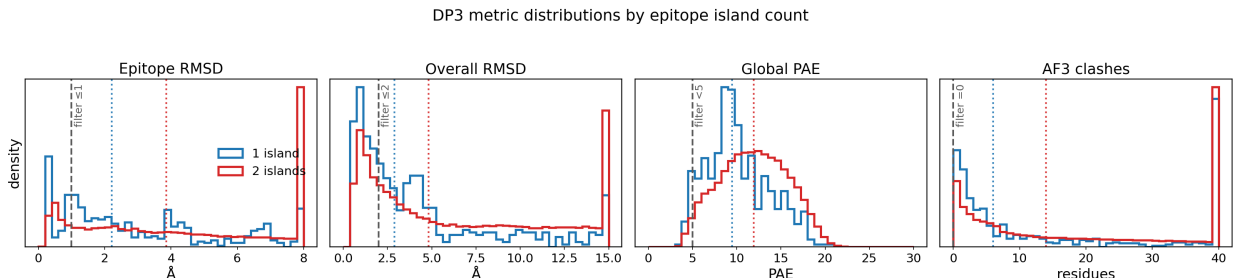


Figure 10: DP3 design metrics (RFD1+MPNN, `dp2.parquet`) split by epitope island count: blue = single-island ($n = 549$), red = two-island ($n = 123\,813$); only designs with an AF3 result are included. Panels, left to right: epitope-chunk RMSD, overall RMSD, global mean PAE, and number of AF3 clashing residues, the four selection metrics. Distributions are area-normalized (density) because the groups differ in size by $\sim 200\times$. Dotted vertical lines mark each group’s median; dashed grey lines mark the four-filter thresholds (epitope RMSD ≤ 1 , overall RMSD ≤ 2 , PAE < 5 , clashes = 0). Single-island designs are shifted toward better values on every metric (median epitope RMSD 2.22 vs 3.86 Å, median PAE 9.5 vs 11.9). Regenerated by `episcaf_analysis/viz/plot_island_split.py`.

9 The production run: per-island scaffolding of dual-island epitopes

The island analysis (Section 8) turns on a question only an experiment can settle: scaffold each island of a dual-island epitope on its own and assay whether it still binds. This run builds those reagents. It is the first component (“individual island”) of the DP4 library outlined in `docs/DP4designIdeas.pdf` — nominally $46 \times 2 \times 5 = 460$ peptides — and one of several arms of DP4 aimed at its Question 1, what makes a minimal epitope.

We define an *island* as one fixed-coordinate A-segment of the RFDiffusion3 contig string. For example the `dp2` contig 15-15/A1-16/57-57/A81-81/15-15 has two islands, A1-16 (16 residues) and A81-81 (1 residue), separated by generated scaffold of the given lengths. A *dual-island epitope* is a ledger id with `epitope_chunks = 2`, i.e. exactly two such segments; the DP3 set contains 46 of

them. We take *island size* to be the A-segment span in residues (not the count of antibody-contact residues within it).

By design, each dual-island epitope contributes its two islands as separate design targets, and we keep the top 5 designs per island under the composite scorer. The island lengths vary widely — from a single residue to 51 — and Table 4 lists both islands of all 46 epitopes so the run is auditable epitope by epitope. One case needs a rule: an island of a single residue cannot be presented on its own. Rather than drop the whole epitope, we simply *skip the size-1 island and still scaffold its partner*. Five islands are skipped on this ground, one each in 2h32_OP, 3q1s_OP, 6o9i_OP, 7a3t_OP, and 7tzh_OP (marked in Table 4); every epitope still contributes at least one island. Of the 92 islands across the 46 epitopes, 87 are scaffolded, giving 87 individual-island targets and, at five designs each, 435 final designs. The per-island target list (both spans, sizes, and the scaffold/skip flag) and Table 4 itself are generated by `episcaf_analysis/dual_island_targets.py` into `results/dual_island_targets.csv` and `manuscript/tables/`, the single reproducible source for these counts.

Each island is scaffolded into a fixed 103-residue construct — the maximum peptide length the PepSeq assay synthesizes. We call the stretches of generated scaffold lying N-terminal and C-terminal to a fixed island its *flanks*; a single-island contig is therefore $\langle \text{N-flank} \rangle / \text{Aa-b} / \langle \text{C-flank} \rangle$, and the scaffold budget the flanks share is $103 - \ell$ residues for an island of length ℓ . We note one unexplained convention difference: Lawson’s original dual-island contigs were all 104 residues, not 103. The change to 103 appears to have been made at some point without a recorded reason; we adopt 103 as the assay ceiling and expect a single terminal residue, averaged over sequence space, to be immaterial, but flag it here because its origin is unknown.

Contig sampling and run size. A *contig* pairs the fixed island(s) with one concrete scaffold-length layout, and each contig is sampled at $8 \text{ RFdiffusion3 backbones} \times 8 \text{ ProteinMPNN sequences} = 64$ designs (the per-contig depth throughout this project). The run size is thus set by how many contigs we generate. We record how Lawson reached his 151 232-design DP3 set, so the scale here is never mysterious: for each epitope he held the two islands fixed and swept the scaffold layout, keeping the two flanks balanced ($|\text{N} - \text{C}| \leq 1$) and stepping the flank length over every integer from 1 to about 20, which makes the *inter-island gap* sweep from ~ 46 down to a floor near 8 (the islands are never allowed to touch). This is a deterministic enumeration, not random sampling: it yields a median of ~ 41 contigs per epitope, so $59 \text{ epitopes} \times 41 \approx 2363$ contigs and $2363 \times 64 = 151\,232$ designs. The quantity his contigs actually vary is the spacing between the two islands.

Our single-island targets have no inter-island gap, so that lever does not exist; the only scaffold freedom is the *flank asymmetry* — where along the 103-mer the island sits, i.e. the split of the $103 - \ell$ budget into N- and C-flank. We therefore generate V contigs per island by *randomly* sampling V distinct N-flank lengths over $[1, 103 - \ell - 1]$ (under a fixed seed, so the ledger is reproducible). This is an explicitly different diversity axis from Lawson’s — scaffold position and topology, not inter-island geometry — which we name here rather than leave implicit. With V variants the run is

$$\underbrace{87}_{\text{islands}} \times V \times \underbrace{8}_{\text{RFD3}} \times \underbrace{8}_{\text{MPNN}} = 5568 V \text{ designs.} \quad (5)$$

We use $V = 20$, giving 1740 contigs and 111 360 designs — the order of the previous DP3 set ($V \approx 27$ would reach its full $\sim 150\,000$). V is the `--variants` dial of `episcaf_pipeline/build_dual_island_designs.py`. Each RFdiffusion3 task emits 8 backbones (`n_batches=1`), so the 8-per-contig depth is one task, not eight: the RFD3 step is 1740 array tasks (one per contig), each yielding 8 backbones that ProteinMPNN then re-sequences 8 ways.

Table 4: Island lengths of the 46 dual-island DP3 epitopes, the input to the per-island design run (Section 9). Each island is one contig A-segment; the value in parentheses is its length in residues. An island marked [†] is a single residue and is skipped (it cannot be presented alone); the epitope’s other island is still scaffolded. Of the 92 islands, 87 are scaffolded (size ≥ 2) and 5 skipped.

epitope	island 1	island 2	designed
2h32_OP	A1--16 (16)	A81--81 (1) [†]	larger only
3q1s_OP	A14--30 (17)	A107--107 (1) [†]	larger only
6o9i_OP	A6--23 (18)	A54--54 (1) [†]	larger only
7a3t_OP	A150--150 (1) [†]	A298--300 (3)	larger only
7tzh_OP	A109--134 (26)	A148--148 (1) [†]	larger only
5eu7_1P	A12--13 (2)	A91--126 (36)	both
5fhx_1P	A14--15 (2)	A62--74 (13)	both
6ztr_OP	A16--17 (2)	A43--63 (21)	both
4u6v_OP	A152--179 (28)	A242--245 (4)	both
5j13_OP	A36--49 (14)	A103--106 (4)	both
6okm_OP	A39--42 (4)	A54--69 (16)	both
6qb6_OP	A40--43 (4)	A143--157 (15)	both
7uxl_1P	A56--80 (25)	A122--125 (4)	both
2xqb_OP	A19--23 (5)	A35--85 (51)	both
4kv5_3P	A28--32 (5)	A90--101 (12)	both
4nnp_OP	A198--210 (13)	A230--234 (5)	both
6ppg_1P	A55--69 (15)	A105--109 (5)	both
5gjs_OP	A12--17 (6)	A29--45 (17)	both
5o4g_OP	A129--134 (6)	A150--169 (20)	both
7so5_OP	A15--27 (13)	A56--61 (6)	both
8cz8_1P	A107--116 (10)	A139--144 (6)	both
2qqn_OP	A21--29 (9)	A45--51 (7)	both
6o16_1P	A52--59 (8)	A100--106 (7)	both
6wgl_OP	A38--47 (10)	A66--72 (7)	both
8oxv_OP	A242--270 (29)	A627--633 (7)	both
3ma9_OP	A19--64 (46)	A186--193 (8)	both
5l6y_OP	A10--17 (8)	A82--93 (12)	both
6cyf_OP	A36--43 (8)	A104--111 (8)	both
7ox3_OP	A6--13 (8)	A63--75 (13)	both
8wsq_OP	A350--371 (22)	A387--394 (8)	both
3ux9_1P	A18--32 (15)	A115--123 (9)	both
4xwo_5P	A197--218 (22)	A249--257 (9)	both
6wmw_1P	A4--25 (22)	A69--77 (9)	both
8db4_OP	A6--18 (13)	A54--62 (9)	both
7kql_OP	A28--49 (22)	A89--98 (10)	both
8jnk_3P	A28--43 (16)	A80--89 (10)	both
6o39_OP	A27--45 (19)	A90--100 (11)	both
6ye3_2P	A56--66 (11)	A79--103 (25)	both
4zfg_OP	A155--175 (21)	A192--203 (12)	both
8pww_OP	A50--61 (12)	A123--137 (15)	both
8trt_1P	A73--94 (22)	A105--117 (13)	both

Table 4 (continued)

epitope	island 1	island 2	designed
8uky_1P	A16--34 (19)	A113--125 (13)	both
6oan_0P	A40--57 (18)	A132--146 (15)	both
5kw9_0P	A118--134 (17)	A152--170 (19)	both
7zxk_1P	A3--19 (17)	A108--130 (23)	both
8t58_0P	A31--60 (30)	A95--120 (26)	both

9.1 One metrics table for every decision

With the run designed above and pushed through RFDiffusion3 $\rightarrow$ ProteinMPNN $\rightarrow$ AlphaFold3, we extract every per-design metric *once* into a single table and make all downstream choices (composite scoring, top-5 per island) from it, so the analysis is never recomputed. This is our analog of Lawson’s `dp2.parquet`: one row per design, written by `scripts/stage05_extract_metrics.py` from the `04_af3/outputs` predictions, the per-island ledger (`results/dual_island_designs.csv`), and the native AbDb complex PDBs.

Why a dedicated extractor. The repository already held two metric extractors, and neither fit this run directly. `compute_metrics.py` is the Lawson-validated antibody extractor, but its run mode is keyed on `dp2`’s hashed tokens and the no-MPNN (RFD3 $\rightarrow$ AF3) layout. `build_12mer_metrics.py` emits exactly the columns the composite scorer reads (`epitope_pae`, the cylinder terms) but parses the 12-mer naming and assumes no antibody. Stage 05 reuses the *validated primitives* of the first (backbone RMSD, PAE, AlphaFold3 file discovery, the clash geometry) and mirrors the PAE decomposition and cylinder calls of the second, but drives them from our per-island ledger.

The per-island set sits in a useful middle: these are the 46 dual-island *mAb* epitopes, so a solved antibody exists and the real DP3 clash filter *is* available. We therefore compute both accessibility signals — the true antibody clash (`af3_n_clash_res`, against native chains B/C) *and* the cylinder surrogate — which makes this run a second check of the surrogate against ground truth, on top of the DP3 check in Section 6.

What each design row carries. The table groups into identity, geometry, confidence, and accessibility (Table 5). Geometry and confidence follow the four-filter definitions of Section 5.1; the accessibility block carries the real clash and the cylinder side by side.

Table 5: Schema of the per-design metrics table (`metrics_dual_island.parquet`). One row per AlphaFold3 design.

column	meaning	role
<i>Identity (which design this is)</i>		
<code>id</code>	epitope id (e.g. 2h32_0P)	select / scope
<code>island_index</code>	which of the two islands (0/1)	select group
<code>island_size</code>	island span (residues)	strata
<code>contig_id</code>	contig within the epitope	provenance
<code>variant, n_flank, c_flank</code>	flank-split placement	provenance
<code>seed, rep, dldesign</code>	RFD3 / MPNN replica indices	provenance

column	meaning	role
predID, af3_dir	design stem, output path	join key
<i>Geometry (backbone RMSD, Kabsch-superposed)</i>		
epitope_chunk_rmsd	design vs AF3, epitope only (≤ 1)	gate + 0.35
overall_rmsd	design vs AF3, full scaffold (≤ 2)	0.15
<i>Confidence (AlphaFold3)</i>		
epitope_pae	mean PAE over epitope residues	0.25
scaffold_pae	mean PAE over the scaffold	diagnostic
mean_pae	global mean PAE (< 5 , four-filter)	diagnostic
ptm, ranking_score	AF3 summary scalars	diagnostic
<i>Accessibility (this run has both)</i>		
af3_n_clash_res	real antibody clash count (= 0)	ground truth
cylinder_ca_clashes	scaffold C α in the cylinder	0.25 (plain)
cylinder_native_aware	same, native volume carved out	surrogate
native_in_cylinder	native non-epitope C α in cylinder	test

Definitional choices (held to the DP3-validated convention). Three choices fix the numbers, and we keep each at the value that makes this set comparable to DP3. (i) Both RMSDs are computed on the *backbone* (N, C α , C, O) and Kabsch-superposed, as in `compute_metrics.py`, not the C α -only variant of the 12-mer extractor. (ii) The cylinder uses `exclude_dist = 1 Å`, the value adopted for the validated 12-mer preset (Section 6). (iii) The epitope is the *whole island span* [`n_flank`, `n_flank + island_size`) — the residues stage 03 actually held FIXED during Protein-MPNN — not the contacts-only subset stored in the ledger. The native-frame and design-frame epitope residues pair one-to-one because each island A-segment is contiguous in both, which we verified holds for all 1740 ledger contigs ($b - a + 1 = \text{island_size}$).

How it is produced. The extractor caches each native complex once per epitope (not once per design) and shards the ~ 46 epitopes across worker processes:

```
python3 scripts/stage05_extract_metrics.py \
  --af3_out_dir runs/dual_island_rfd3/04_af3/outputs \
  --mpnn_pdb_dir runs/dual_island_rfd3/03_mpnn/mpnn_pdbs \
  --ledger      results/dual_island_designs.csv \
  --native_dir  <AbDb cleaned complex dir> \
  --out         runs/dual_island_rfd3/05_analysis/metrics_dual_island.parquet \
  --workers 16
```

It writes the parquet and a CSV alongside. A companion report, `scripts/stage05_summarize.py`, prints the breakdowns below and writes the per-island summary `results/dual_island_gate_summary.csv` (so these numbers are regenerable, not pasted).

9.2 What the run yielded

The full run scored 111 322 designs across all 46 epitopes and 87 islands, every one with a status of ok. The designs are good where it matters: median epitope-chunk RMSD is 1.20 Å (best 0.03 Å)

and median epitope PAE is 3.1, while the scaffold is predicted less confidently (scaffold PAE median 7.9) — the same epitope-sharp / scaffold-loose split the decomposed metrics were built to exploit (Section 5.2).

The design funnel. Table 6 walks a design through each filter. Two thirds clear the composite scorer’s gate, and nearly half hold the epitope to within 1 Å. The strict DP3 four-filter, however, passes only 0.37% — driven down by the global PAE (< 5) and the zero-clash requirements, exactly the regime seen on DP3 ($< 1\%$, Section 8). This is why per-island *selection ranks by the composite score within the gate*, rather than taking only four-filter passes, which would leave most islands with nothing to advance.

filter (status = ok, $n = 111\,322$)	designs passing	%
composite scorer gate (epitope RMSD ≤ 2.5 Å)	74 061	66.5
epitope-chunk RMSD ≤ 1 Å	51 653	46.4
overall RMSD ≤ 2 Å	45 531	40.9
global mean PAE < 5	25 849	23.2
antibody clash = 0 (of 107 482 with clash)	2510	2.3
DP3 four-filter, all four (real clash)	407	0.37

Table 6: Per-design pass counts. Indented rows are the individual four-filter thresholds (not nested); the last row requires all four at once. Regenerated by `scripts/stage05_summarize.py`.

These are single-island rates, and the comparison to Lawson’s DP3 is two-edged. It is tempting to read Table 6 against the four-filter rates of Lawson’s DP3 set, but the two differ in *two* ways at once: the backbone generator (RFD1 vs RFD3) *and* the epitope (Lawson’s set is dominated by two-island epitopes; this run scaffolds single islands). Table 7 holds each fixed in turn, over the same 59 mAb epitopes. On the *same* DP3 contigs, swapping RFD1 for RFD3 barely moves the rates — and slightly lowers most — so the generator is not the story (the full distributions are in Section 4). Scaffolding a *single* island is: the epitope-RMSD pass rate roughly triples ($14\% \rightarrow 46\%$) and the global-PAE rate climbs from 7% to 23%, because one small, contiguous island is far easier to hold rigidly and predict confidently than two spatially separated patches (Section 8). The one filter that moves the other way is accessibility: the clash-free rate *falls* ($8\% \rightarrow 4\% \rightarrow 2\%$ over all designs), because a small epitope on a fixed 103-mer leaves more non-epitope scaffold mass to intrude on the antibody approach — the scaffold-mass effect of Section 6. That cap on clash is why the all-four rate (0.37%) lands *below* Lawson’s (0.56% over all designs, 0.68% over its valid ones) even though three of the four filters improve markedly.

filter	RFD1 / Lawson* DP3 contigs	RFD3 / same DP3 epitopes	RFD3 / single island (this run)
epitope RMSD ≤ 1 Å	13.7 (16.6)	14.0	46.4
overall RMSD ≤ 2 Å	22.2 (26.9)	16.4	40.9
clash = 0	8.3 (10.1)	3.9	2.3
mean PAE < 5	1.9 (2.3)	7.1	23.2
all four	0.56 (0.68)	0.50	0.37

Table 7: Per-filter pass rates (%) disentangling generator from island count, same 59 mAb epitopes throughout. Column 1→2 isolates RFD1 vs RFD3 (identical contigs); column 2→3 isolates two-island vs single-island (both RFD3). Each cell is the rate over *all* designs generated. *The RFD1 parenthetical is the rate over only its *valid* designs (all four metrics present); the two diverge because Lawson’s set is missing an AlphaFold3 result for 26 870 of its 151 232 designs (its valid column reproduces the reference hand calculation, 17/27/10/2/0.7). The RFD3 columns show a single rate because those sets are essentially complete (RFD3/same missing 12 of 150 948; RFD3/single missing only a clash for the 3840 crystal-gap designs of 111 322), so their all- and valid-rates coincide. Regenerated by `scripts/compare_passrates.py`.

Top-5 per island is reachable almost everywhere. Ranking within the gate, **83 of the 87 islands have at least five designs to advance**. The four that fall short (Table 8) are *all large islands, 21–28 residues* — the island-size difficulty of Section 8 surfacing directly in the deliverable: the biggest islands are the hardest to scaffold cleanly, and three of them produce no gated design at all. For these we advance whatever clears the gate (fewer than five, or none) and say so, rather than padding the set with designs that miss the bar.

epitope	island	island size (aa)	gated designs (of 1280)
4u6v_OP	0	28	0
4zfg_OP	0	21	0
7uxl_1P	0	25	0
8t58_OP	1	26	1

Table 8: The four islands short of a full top-5 (gate epitope RMSD ≤ 2.5 Å); all are large. Full per-island counts in `results/dual_island_gate_summary.csv`.

Where accessibility is unavailable: crystal gaps. For 3840 designs — one island each of three epitopes (5eu7_1P, 5fhx_1P, 6ztr_OP, 1280 designs apiece) — the clash and cylinder columns are null, with status `too_few_epitope_pairs`. The cause is the same crystal coverage constraint as in Section 6: those islands’ residues are unresolved in the native complex, so there are no native C α anchors to superimpose the design onto, and neither the real antibody clash nor the cylinder can be placed. These designs keep their geometry and PAE (and so are still gated and ranked on those terms); they simply carry no accessibility score.

9.3 Selection

The composite scorer (Section 5.4) reads this table directly. For the per-island deliverable we select the top five per (`id`, `island_index`) — per *island*, not per epitope — so each island is ranked on its own and the PepSeq assay can deconvolve which island drives binding. The policy is to **rank, not gate**: we apply no hard threshold and take the top five designs per island by composite score, since the composite already penalizes weak metrics softly and a hard cut would only discard designs the ranking buries anyway. Because this is a known-antibody set, the accessibility term in the composite is the *real* clash `af3_n_clash_res` rather than the cylinder surrogate (Section 5.4): a design that places mass where the antibody binds is pushed down the ranking, but is not categorically excluded. [final top-5 list to be regenerated from the locked policy]

10 The DP4 library: what we are shipping

This section inventories the components we contribute to the DP4 assay run and what each one is for. DP4 as a whole is the larger fourth PepSeq library; the pieces assembled here are the designed and control constructs this project adds to it. Each component targets one of the three questions from Section 1, and all of them are emitted in the same constant-length construct (Section 3) and the same DP2 annotated format, so they can be merged into one ordered synthesis file (Section 10.7).

component	what it is	constructs	status
C1 known antibody, whole epitope	full-epitope scaffolds, top-5/epitope	TBD	selection pending
C2 known antibody, single island	87 island contigs, each island alone	~435	generated; selection pending
C3 scaffolds, polyclonal tiling	top-5/window, no antibody	~2181	scored, top-5 selected
C4 linear tiled-30mer controls	bare tiled peptides	2174	built
C5 metric-space sampling	designs spread across metrics	TBD	planned
C6 island alanine mutants	island1/2 $\rightarrow$ Ala, best-20/mAb	~2000	spec; to build

Table 9: The DP4 components assembled here. C1 and C2 are the two halves of the islands experiment (whole epitope vs. one island at a time); C6 tests the same question by mutation. Counts are current best estimates; C1/C2 settle once selection is finalized, C5 once a peptide budget is set. C3 is the 951+450+780 top-5 designs of Table 1; C4 is the 2174 tiles of Section 3.

10.1 C1 — known-antibody epitopes, whole

For each mAb epitope we ship the best few scaffolds that present the *complete* epitope — all of its islands held together in their native geometry, the way the antibody actually sees it. These are the designs whose binding we can score against a ground truth, since the antibody is known; they continue the DP3 set that gave us the binding data of Section 7 and are the comparator for the single-island arm below. Selection ranks on the re-weighted composite with no hard gate (Section 9.3); because the antibody is known, the accessibility term is the *real* clash `af3_n_clash_res` rather than the cylinder surrogate. Whole-epitope presentation is straightforward for the 13 single-island epitopes but hard for multi-island ones — preserving the relative geometry of separated islands is exactly where scaffolding struggles (Section 1, the 2XQB failure) — which is the motivation for shipping C2 alongside it.

10.2 C2 — known-antibody epitopes, one island at a time

The single-island component scaffolds each island of the 46 multi-island epitopes *individually*: one island per design, presented alone. This is the direct test of Q1 (Section 8) — does an antibody need both islands of its epitope, or does one suffice? Shipping C1 (whole epitope) and C2 (each island alone) for the same antibodies lets the assay deconvolve which island, if any, carries the binding. The per-island production run (Section 9) covers 87 scaffolded islands (5 size-1 islands are skipped as un-presentable); at top-5 per island that is on the order of 435 constructs (Table 4). Selection is identical to C1 (composite, no gate, real-clash accessibility), ranked per (`id`, island) so each island competes on its own.

10.3 C3 — scaffolded epitopes from polyclonal tiling

For antigens with no known antibody we tile the whole protein into overlapping windows and scaffold each one (Section 3), then keep the top 5 designs per window by the composite. This is the

polyclonal arm (Q3): we do not know where the real epitope is, so we cover the antigen densely and let the assay find it. This is the one component where the antibody-clash filter cannot be computed, so accessibility is judged entirely by the native-aware cylinder surrogate (Section 6) — the metric the DP3 binding data singled out as the strongest predictor (Section 7.4), which is why it now carries the most weight in selecting these designs. Across the three tiled antigens (1D2K, 6M0J, 4WAT) this is 951+450+780 top-5 designs (Table 1).

10.4 C4 — linear tiled-30mer controls

The linear controls are the bare epitope peptides the scaffolded designs are meant to beat: unscaffolded 30-mers (model (none)) tiled across the antigen, presenting the sequence with none of the conformational constraint a scaffold imposes. They are the baseline for the central scaffold-versus-linear question (Section 1): if conformational presentation matters, a scaffolded design should bind its antibody more reliably than the corresponding linear tile. We tile the gap-free PDB FASTA chain at a 30-residue window, step 6 (Section 3), giving 2174 tiles over 62 antigens (`data/libraries/dp4_tiled30mers_fasta.csv`). This component is built.

10.5 C5 — metric-space sampling

The DP3 binding data is a weak prior precisely because every assayed design already passed the four filters, so the metrics barely varied (Section 7). C5 fixes that by design: instead of shipping only the top designs, we deliberately sample designs *spread across* the metric space — including ones the current filters would reject — so that epitope RMSD, accessibility, and PAE each take a wide range of values. Here accessibility is the cylinder (not the real clash): the point of this component is to learn the predictors that transfer to the no-antibody setting, so we sample along the same surrogate we will have to rely on there. Reading binding off this spread is what lets us learn which metrics actually predict enrichment and finally fit the composite’s weights (Q2, Section 11), rather than re-deriving the filters we started from. The sampling design (a farthest-point spread over the standardized metrics) is worked out; the size is set by the peptide budget, so the count is open.

10.6 C6 — island alanine mutants

For the best 20 designs of each of the 57 mAbs we alanine-scan the epitope islands: every residue of island 1 mutated to alanine, and (where the epitope has a second island) every residue of island 2. The unmutated designs are not part of this component — they are already shipped as C1 — so C6 is the controlled comparison against them. Mutating an island to alanine strips its side-chain chemistry while leaving the fold intact, so if binding drops when island 1 is alanine-substituted the antibody requires island 1, and likewise for island 2. This is the mutation-based read of the islands question (Q1), complementary to C2’s presentation-based read (each island scaffolded alone): C2 asks whether one island *suffices*, C6 asks whether each island is *necessary*. At 20 designs $\times$ 57 mAbs, with one island-mutant for single-island epitopes and two for multi-island ones, this is on the order of 2000 constructs.

Scaffold-mutation control (under discussion). A companion negative control would mutate four 6-residue windows of the *scaffold* (not the epitope) in each design: if binding is epitope-driven, perturbing the scaffold should leave it intact, so a design that survives this but collapses when an epitope island is alanine-substituted is behaving exactly as intended. The open question is how the four windows are chosen. Purely random placement is risky — a window landing on

fold-critical residues could unfold the protein and lose epitope presentation, which would confound the control rather than serve it. We would instead fix the windows *deterministically* and *epitope-avoiding* (surface positions away from the core), so the perturbation tests scaffold sequence without destabilizing the fold, and so the construct set is reproducible. The mutant residue (alanine, or a fold-tolerant choice) is likewise to be decided.

Still to confirm for C6: that “best 20” is the top 20 per mAb by the composite, and the scaffold-control specifics above.

10.7 Assembly into one synthesis file

All six components share the constant 103-mer construct and the eight-column DP2 annotated format (Section 3), so the final step concatenates them into a single ordered library file with global `library_member` numbering — a named, documented pipeline stage (`06_library`) so the shipped file is regenerable from its components. That assembly is the last thing to run once C1/C2 selection is locked and C5’s budget is set.

The assembled peptide library is then encoded to DNA oligos for synthesis with the LadnerLab `oligo_encoding` tools (<https://github.com/LadnerLab/Library-Design>), using the same updated codon-weight table as DP3 (`/tgen_labs/Immunology/ekelley/DP3_PepSeq_Library_Design/new_codon_`). The encoding is the final synthesis-prep step; it is external tooling, so the pipeline ends at the DP2-format peptide file and hands off to it.

11 Open questions and next steps

1. **Within-DP3 island stratification (the clean control).** We have split by island count (Section 8): single-island designs are distributionally easier on every metric. What remains is to split by island *size*, since the single-island epitopes are also shorter on average, to separate the “contiguous-12 is easier” confound from a real scaffolding-difficulty effect. The per-epitope island metadata is in `dp2.parquet` (`epitope_chunks`, `contig_string`).
2. **Move to a sigmoid transform (planned).** The composite currently maps each metric to a percentile within its population, which is non-differentiable and population-dependent, so it cannot be fit and a score does not transfer from one set to another. A sigmoid (fixed midpoint and slope k per metric) gives an absolute, saturating, differentiable score, which both lets the weights be learned and lets a DP3-trained score transfer to DP4. The scorer already supports it (`transform=sigmoid` in `presets.py`); midpoints can be seeded from the DP3 separations (e.g. a cylinder midpoint near 15, between the clash-free and clashing medians). This is the prerequisite for fitting; the percentile version stays the default until then.
3. **Fit the weights once the label has breadth.** The composite is a single-layer perceptron with hand-set weights (feed-forward, no backprop), and the natural next step is to learn them. The first experimental label has now arrived (the DP3 binding data, Section 7): scaffolded designs do enrich over the no-antibody baseline, so the premise holds. But that set cannot yet fix the weights — every assayed design already passed the four filters (no failures to contrast), its pooled metric–binding correlation is a between-antibody artifact, and only two of the eight antibodies are sampled deeply. We therefore still defer the fit, and the requirement for DP4 is specific: assay designs spread across the metric space, *including ones the current filters reject*, so that binding can be regressed on the metrics with real variance and across many epitopes. The cylinder weight in particular is already justified by data twice over: on DP3 it

predicts the true antibody clash at AUC 0.93 (Section 6), and it is the single best predictor of *experimental* binding, the only metric whose correlation survives the between-antibody confound (Section 7.4). So it earns real weight even before a global fit.

4. **Confirm the canonical DP3 metrics builder.** Determine whether `compute_metrics_mpnn.py` (currently archived) is the live RFD3+MPNN metrics producer, and promote it into `episcaf_analysis/` if so (see `docs/MIGRATION.md`).
5. **Validate the DP3 scoring path** end-to-end with `--preset antibody`, confirming `cylinder_ca_clashes` is the correct accessibility column there.
6. **Finish the 12-mer AlphaFold3 data migration** and verify landing counts against the manifest.
7. **Pin down the crystal-gap dropouts.** We can already count them locally: 6M0J and 4WAT lose 15 and 40 tiles to unresolved crystal regions (Table 1). What is left is to confirm these against the cluster pre-filter status (the explicit `crystal_epi_incomplete` entries) and to check whether the lost tiles fall in contiguous runs, i.e. whether whole regions go un-scorable from gaps.

References